Exercise 2 — Concepts

Advanced System Theory · Week 02: System Properties and Linearization

Course: Advanced System Theory (Paderborn University, Automatic Control Group RAT, Prof. Dr. Erdal Kayacan), winter semester 2026/27 **Lecture:** Week 02 — System properties and linearization of nonlinear systems **Covers:** everything needed for Exercise 2 — Task 1 (system properties), Task 2 (equilibria and linearization), Task 3 (evolution of a discrete-time system, internal and BIBO stability)

This sheet builds every idea from the ground up and assumes only school calculus and a little linear algebra. Each test is given as a recipe you can apply mechanically, followed by worked examples. The full solutions are in `solutions.md`; the scripts in `python/` check every number.

Notation (as in the lecture slides).

- As on slide 5: **bold symbols are non-scalar quantities** (vectors and matrices): state $\boldsymbol{x} \in \mathbb{R}^n$, input $\boldsymbol{u} \in \mathbb{R}^m$, output $\boldsymbol{y}$, vector field $\boldsymbol{f}$, output map $\boldsymbol{g}$, system matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$. Plain symbols are scalars: components x_1, x_2, scalar signals $u(t), y(t)$ (as in Task 1), numbers $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$.
- $y(\boldsymbol{x}_0, \boldsymbol{u}, t)$ is the output at time t for initial state $\boldsymbol{x}(t_0) = \boldsymbol{x}_0$ and input $\boldsymbol{u}$ (Definitions 1.1, 1.2). $u_1(t) \to y_1(t)$ reads "input u_1 produces output y_1" (slide 12).
- $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \dot{\boldsymbol{x}}$. Operating/equilibrium point $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ with deviations $\Delta\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$, $\Delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_0$, $\Delta\boldsymbol{y} = \boldsymbol{y} - \boldsymbol{y}_0$; $\cdot|_0$ means evaluation at $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ (slide 25). Task 2(c) of the sheet writes the operating point as $(\boldsymbol{x}^*, u^*)$.
- Continuous time $t \in \mathbb{R}$ with round brackets $x(t)$; discrete time $k, n \in \mathbb{Z}$ with square brackets $\boldsymbol{x}[k]$, $y[n]$.

Contents

1. Signals, systems and how to manipulate signals
2. Linearity
3. Time-invariance
4. Causality
5. Invertibility and stability
6. Worked lecture exercise: the property table
7. Nonlinear state-space models
8. Equilibrium points and how to find them
9. Linearization with Jacobians
10. What the linear model says about stability (2×2 toolkit)
11. Discrete-time LTI systems: solution of the state equation
12. Computing $\boldsymbol{A}^k$ with eigenvalues and eigenvectors
13. Modes, internal stability and BIBO stability
14. Cheat sheet

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1. Signals, systems and how to manipulate signals

1.1 What a "system" is here

A system maps an input signal to an output signal. With the arrow notation of slide 12,

$$u(t) \;\rightarrow\; y(t),$$

and, when the system has a state, the output also depends on the initial state: $y(\boldsymbol{x}_0, \boldsymbol{u}, t)$ (slide 5). The property definitions compare **whole signals** — the values at all times t — not single values.

In Task 1 the systems are given as explicit **I/O relationships** $y(t) = \ldots$ with scalar u and y. They have no initial state, so the argument $\boldsymbol{x}_0$ is simply dropped. Some are **memoryless (static)**: $y(t)$ depends only on u at the same instant t (e.g. $y(t) = u(t)\cos(2\pi f_0 t + \phi)$). Others have **memory**: $y(t)$ depends on u at other times (e.g. $u(t+1)$ or $u(2t)$).

In Tasks 2 and 3 the systems are given as **state-space models** — differential or difference equations with a state $\boldsymbol{x}$ (§7, §11).

1.2 Time shift, time scaling, and the "replace t" rule

Three operations appear constantly. Let $u(t)$ be a signal and $T > 0$.

Operation	New signal	Picture
Delay by T	$u(t-T)$	graph moves **right** by T (events happen later)
Advance by T	$u(t+T)$	graph moves **left** by T (events happen earlier)
Compression by 2	$u(2t)$	graph squeezed towards $t = 0$; runs twice as fast

Why does $u(t-T)$ move right? The value that u had at time 0 is now found where $t - T = 0$, i.e. at $t = T$.

The "replace t" rule (the single most important trick for Task 1). If $y(t) = u(2t)$ and you want the output at another time, say $y(t-T)$, you replace *every* t in the formula by $t - T$:

$$y(t-T) = u\big(2(t-T)\big) = u(2t-2T).$$

The common mistake is to write $u(2t-T)$. That is a different signal: it is u first delayed by T and *then* compressed. Keep the two orders strictly apart:

- **Delay the input first, then apply the system:** feed the delayed input $u(t-T)$ into the system. In $y(t) = u(2t)$ the system reads its input at time $2t$, so the output is $u(2t-T)$. In the lecture notation this is $y(u(t-T), t)$.
- **Apply the system first, then delay the output:** $y(t-T) = u(2(t-T)) = u(2t-2T)$. In the lecture notation this is $y(u(t), t-T)$.

The safest way to avoid errors is to give the delayed input its own name ($\tilde{u}(t) := u(t-T)$) and only then substitute.

1.3 How properties are proved and disproved

All properties in this sheet are "for all" statements: *for all* inputs, *for all* shifts $T > 0$, *for all* $\alpha_1, \alpha_2 \in \mathbb{R}$.

- To **prove** a property you must show it in general (with symbols, not numbers).
- To **disprove** it, a **single counterexample** is enough: one particular input, one particular T, one particular α for which it fails.

This is why the solutions show general derivations for "Y" and one concrete counterexample for "N".

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2. Linearity

2.1 Definition 1.1 — the superposition principle

For a linear system, the following conditions for $t > t_0$ and $\alpha_1, \alpha_2, \beta_1, \beta_2 \in \mathbb{R}$ are fulfilled for all (permissible) initial states $\boldsymbol{x}(t_0) = \boldsymbol{x}_0$, input signals $\boldsymbol{u}(t)$ and initial time $t_0 > 0$:

$$\begin{gathered}\boldsymbol{y}(0,\ \alpha_1\boldsymbol{u}_1 + \alpha_2\boldsymbol{u}_2,\ t) = \alpha_1\boldsymbol{y}(0, \boldsymbol{u}_1, t) + \alpha_2\boldsymbol{y}(0, \boldsymbol{u}_2, t),\\ \boldsymbol{y}(\beta_1\boldsymbol{x}_{0,1} + \beta_2\boldsymbol{x}_{0,2},\ 0,\ t) = \beta_1\boldsymbol{y}(\boldsymbol{x}_{0,1}, 0, t) + \beta_2\boldsymbol{y}(\boldsymbol{x}_{0,2}, 0, t),\\ \boldsymbol{y}(\boldsymbol{x}_0, \boldsymbol{u}, t) = \boldsymbol{y}(0, \boldsymbol{u}, t) + \boldsymbol{y}(\boldsymbol{x}_0, 0, t).\end{gathered}$$

Line by line:

1. linear in the **input** (with zero initial state);
2. linear in the **initial state** (with zero input);
3. total response = **zero-state (forced) response** + **zero-input (free) response**.

The superposition principle comprises **additivity** (the response to a sum is the sum of the responses) and **homogeneity** (scaling the input scales the output) with respect to the initial state and the input. You will see the split of line 3 exactly in Task 3(c): $\boldsymbol{x}[k] = \underbrace{\boldsymbol{A}^k\boldsymbol{x}[0]}_{\text{free}} + \underbrace{\sum \boldsymbol{A}^{k-1-i}\boldsymbol{B}u[i]}_{\text{forced}}$.

2.2 The form used for I/O relationships (slide 12)

Without an initial state only line 1 remains. The lecture writes it with arrows:

$$\text{CT:}\quad x_1(t) \to y_1(t) \ \text{ and } \ x_2(t) \to y_2(t) \implies a\,x_1(t) + b\,x_2(t) \ \to \ a\,y_1(t) + b\,y_2(t),$$

$$\text{DT:}\quad x_1[n] \to y_1[n] \ \text{ and } \ x_2[n] \to y_2[n] \implies a\,x_1[n] + b\,x_2[n] \ \to \ a\,y_1[n] + b\,y_2[n].$$

(On slides 11–13 the input is called x; in Task 1 it is called u.) In Task 1 notation, with α_1, α_2 as in Definition 1.1:

$$u_1(t) \to y_1(t),\ \ u_2(t) \to y_2(t) \quad \implies \quad \alpha_1 u_1(t) + \alpha_2 u_2(t) \ \to \ \alpha_1 y_1(t) + \alpha_2 y_2(t).$$

Every LTI state-space model $\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t)$, $\boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t)$ is linear in the sense of Definition 1.1.

2.3 Recipe: testing an I/O relationship for linearity

1. **Quick necessary test — zero in, zero out.** Put $u(t) = 0$ for all t. A linear system must give $y(t) = 0$ for all t, because homogeneity with $\alpha_1 = 0$ gives $0 \cdot u_1 \to 0 \cdot y_1 = 0$. If $y \not\equiv 0$, the system is **not linear** — done.
2. **General test.** Let $u_1(t) \to y_1(t)$, $u_2(t) \to y_2(t)$ and define $u_3(t) = \alpha_1 u_1(t) + \alpha_2 u_2(t)$. Compute its output $y_3(t)$ by inserting u_3 into the formula wherever u appears (with the *same* time argument). Check whether $y_3(t) = \alpha_1 y_1(t) + \alpha_2 y_2(t)$.
3. **If they differ for some choice**, give one concrete counterexample.

2.4 Typical patterns

Relationship	Linear?	Reason
$y(t) = c(t)\, u(t-\tau)$, any known function $c(t)$	Y	u appears to the first power, multiplied only by things that do not depend on u
$y(t) = u(\psi(t))$, any time map ψ (e.g. $u(2t)$, $u(-t)$)	Y	changing the *time argument* is linear: $u_3(\psi(t)) = \alpha_1 u_1(\psi(t)) + \alpha_2 u_2(\psi(t))$
sums, integrals, derivatives of u	Y	these operations are linear
$y = u^2$, $3x(t)x(t-1)$, $\sqrt{u}$, $\lvert u\rvert$	N	not homogeneous: $(2u)^2 = 4u^2 \neq 2u^2$
$y = \cos(u)$, $\sin(\cdots + u)$, e^u	N	nonlinear function *of the input value*
$y = 3u + 1$ (affine)	N	fails zero-in-zero-out

Key distinction (needed for Task 1 c, d): a function of *time* that multiplies the input, like $u(t)\cos(2\pi f_0 t + \phi)$, keeps the system linear. A nonlinear function *applied to the input value*, like $A\cos(2\pi f_0 t + u(t))$, destroys linearity.

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3. Time-invariance

3.1 Definition 1.2

A system is called **time-invariant** if, for every admissible initial state $\boldsymbol{x}_0$, every admissible input $\boldsymbol{u}(t)$, and every time shift $T > 0$,

$$\boldsymbol{y}(\boldsymbol{x}_0,\ \boldsymbol{u}(t),\ t - T) = \boldsymbol{y}(\boldsymbol{x}_0,\ \boldsymbol{u}(t-T),\ t).$$

- **Left side:** the original output, observed T later — i.e. the output delayed by T.
- **Right side:** the output at time t when the input is delayed by T.

In words: *the behaviour and characteristics of the system are fixed over time* — the same experiment carried out later gives the same result, just later. Otherwise the system is **time-variant**. In the lecture's solution (slide 12): a time shift of the input signal results in the same time shift of the output signal.

A system is **linear and time-invariant (LTI)** if the system matrices $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ are constant ($\boldsymbol{A}(t) = \boldsymbol{A}\ \forall\, t > 0$ and so on):

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{A}\boldsymbol{x}(t) + \boldsymbol{B}\boldsymbol{u}(t), \qquad \boldsymbol{y}(t) = \boldsymbol{C}\boldsymbol{x}(t) + \boldsymbol{D}\boldsymbol{u}(t).$$

3.2 Recipe: compare both sides of Definition 1.2

For an I/O relationship $y(t) = \ldots$ (no $\boldsymbol{x}_0$):

1. **Right side $y(u(t-T),t)$ — delay the input, then apply the system.** In the formula, replace every $u(\text{something})$ by $u(\text{something} - T)$. Leave every other t untouched.
2. **Left side $y(u(t),t-T)$ — apply the system, then delay the output.** Take the original formula $y(t)$ and replace **every** t by $t-T$ (the "replace t" rule of §1.2).
3. **Compare.** If both sides agree for all u, t and $T > 0$, the system is time-invariant; otherwise it is time-variant.

3.3 Red flags and patterns

- t **appears explicitly outside the input** (as in $\cos(2\pi f_0 t)$, $t\,u(t)$, $0.5e^t u(t)$): almost always time-variant. On the right side the explicit t stays t; on the left side it becomes $t-T$. *Exception:* if the explicit time function is constant (e.g. $f_0 = 0$), it disappears and the system can be time-invariant.
- **Time scaling or reversal of the input** ($u(2t)$, $x[4-n]$): time-variant, because a shift T of the input becomes a shift of $T/2$ (or $-T$) at the output.
- **Pure shifts of the input** ($u(t-1)$, $u(t+1)$): time-invariant.
- **State space:** constant $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ ⇒ time-invariant; $\boldsymbol{A}(t)$, ... depending on time ⇒ generally time-variant.

Example (time-invariant). $y(t) = u(t-1)$. Right side: $y(u(t-T),t) = u(t-1-T)$. Left side: $y(u(t),t-T) = u(t-T-1)$. Equal ⇒ time-invariant.

Example (time-variant). $y(t) = t\,u(t)$. Right side: $y(u(t-T),t) = t\,u(t-T)$. Left side: $y(u(t),t-T) = (t-T)\,u(t-T)$. These differ (e.g. $u(t) = 1$, $T = 1$: t vs $t-1$) ⇒ time-variant.

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4. Causality

4.1 Definition 1.3

A system is called **causal** if and only if for any input signal pair $u_1(t)$ and $u_2(t)$ and any $T \in \{\mathbb{R} \mid 0 < T\}$ such that

$$u_1(t) = u_2(t), \quad \forall t \leq T,$$

the corresponding outputs satisfy

$$y_1(t) = y_2(t), \quad \forall t \leq T.$$

In simple words: **a system is causal if the output depends on past and current inputs but not on future inputs.** Every physical system that runs in real time is causal: it cannot react to something that has not happened yet.

4.2 Recipe

For the output at time t, list every time instant at which the input is read ("read times").

- If **all** read times are $\leq t$ **for all** t ⇒ **causal**.
- If **some** read time is $> t$ for **some** t ⇒ **non-causal**. Counterexample: take u_1, u_2 that agree for $t \leq T$ but differ just after T (Definition 1.3 with that T).

Memoryless systems read only time t, so **every memoryless system is causal**.

4.3 Patterns

Read time	Causal?	Comment
$u(t)$, $x(t-1)$, $\sum_{k=-\infty}^{n} x[k]$	Y	present and past only
$u(t+1)$	N	needs the input one second in the future
$u(2t)$	N	for $t > 0$: $2t > t$, i.e. the future (for $t < 0$ it reads the past, but causality must hold for *all* t)
$x[4-n]$	N	for $n = 0$ it reads $x[4]$, the future

Non-causal does not mean useless: in offline signal processing (a recorded file) the "future" samples are available. It only means the system cannot run in real time.

4.4 State-space models are causal

$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ is integrated forwards: $\boldsymbol{x}(t) = \boldsymbol{x}(t_0) + \int_{t_0}^{t} \boldsymbol{f}(\boldsymbol{x}(\tau), \boldsymbol{u}(\tau))\,\mathrm{d}\tau$ uses inputs only up to t. As the lecture states, **the LTI state-space representation is causal by definition.**

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5. Invertibility and stability

These two are not asked for in Task 1, but they are part of the lecture and of the lecture exercise in §6.

5.1 Definition 1.4 — invertible system

A system is said to be **invertible** if distinct inputs lead to distinct outputs.

- $y(t) = 2u(t)$ has the inverse system $w(t) = 0.5\,y(t)$ (slide 8).
- $y(t) = u(t)^2$ is not invertible (u and $-u$ give the same output).
- $y(t) = u(t)\cos(2\pi f_0 t + \phi)$ is not invertible: at the zeros of the cosine the input value is lost.

5.2 Definition 1.5 — stability

A system is said to be **stable** if small perturbations to the system lead to responses that do not diverge. In layman's terms (slide 10): a stable system tends to return to a particular state after being temporarily disturbed (ball in a valley); an unstable one moves away (ball on a hilltop).

Two precise versions are used in the course:

- **BIBO stability (external, week 10)** — used in the lecture's solution (slide 12): every **b**ounded **i**nput produces a **b**ounded **o**utput: $|u(t)| \le M_u < \infty\ \forall t \ \Rightarrow\ |y(t)| \le M_y < \infty\ \forall t$.
- **Internal stability (week 9)** — with zero input, the state returns to (or stays near) the equilibrium after a perturbation of the initial state. For linear systems this is decided by the eigenvalues of $\boldsymbol{A}$ (§10, §13).

Lecture question (slide 9): $y(t) = 0.5e^t u(t)$. The bounded input $u(t) = 1$ gives $y(t) = 0.5e^t \to \infty$. Not BIBO stable.

BIBO recipe for I/O relationships. Assume $|u| \le M$. Try to bound $|y|$ using $|\cos| \le 1$, the triangle inequality $|a+b| \le |a| + |b|$, etc. If a bound exists ⇒ BIBO stable. If you find one bounded input with an unbounded output (sums that grow, factors like e^t or t) ⇒ not BIBO stable.

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6. Worked lecture exercise: the property table

Slides 11–13 ask for causality, stability, time-invariance and linearity of four systems (input x, output y). Working them through is the best preparation for Task 1.

(1) $y[n] = x[n-4] + x[4-n]$

- *Causal?* Read times $n-4$ and $4-n$. For $n=0$: $4-n = 4 > 0$ (future). **N.**
- *Stable?* $|y[n]| \le |x[n-4]| + |x[4-n]| \le 2M$. **Y.**
- *Time-invariant?* Input delayed: $x[n-4-T] + x[4-n-T]$. Output delayed: $y[n-T] = x[n-T-4] + x[4-n+T]$. The second terms differ ($-T$ vs $+T$) because of the time reversal. **N.**
- *Linear?* Only time arguments are changed. **Y.**

(2) $y(t) = 3x(t) + x(t-1)$ — present and past ⇒ causal; $|y| \le 4M$ ⇒ stable; pure shifts with constant coefficients ⇒ time-invariant; first powers only ⇒ linear. **Y Y Y Y.**

(3) $y(t) = 3x(t)x(t-1)$ — causal, stable ($|y| \le 3M^2$), time-invariant; but the input $2x$ gives $3(2x(t))(2x(t-1)) = 4 \cdot 3x(t)x(t-1) \ne 2\,y(t)$ ⇒ **not linear**. **Y Y Y N.**

(4) $y[n] = \sum_{k=-\infty}^{n} x[k]$ **(accumulator)** — past and present ⇒ causal. The bounded input $x[k] = 1$ for $k \ge 0$ (0 before) gives $y[n] = n+1 \to \infty$ ⇒ **not stable**. Input delayed: $\sum_{k\le n} x[k-T] = \sum_{j\le n-T} x[j]$; output delayed: $y[n-T] = \sum_{k\le n-T} x[k]$; equal ⇒ time-invariant. A sum is linear ⇒ linear. **Y N Y Y.**

	causal	stable	time-invariant	linear
$y[n] = x[n-4] + x[4-n]$	N	Y	N	Y
$y(t) = 3x(t) + x(t-1)$	Y	Y	Y	Y
$y(t) = 3x(t)x(t-1)$	Y	Y	Y	N
$y[n] = \sum_{k=-\infty}^{n} x[k]$	Y	N	Y	Y

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7. Nonlinear state-space models

7.1 The general form (slide 16)

A nonlinear but time-invariant dynamic system can usually be represented by a set of nonlinear differential equations in the form

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}),$$

where $\boldsymbol{f} : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$ is a continuously differentiable nonlinear vector field, $\boldsymbol{x} \in \mathbb{R}^n$ is the state vector and $\boldsymbol{u} \in \mathbb{R}^m$ is the input vector. If there is an output, it is given by a time-invariant output equation $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u})$ (slide 22). Written out for $n = 2$:

$$\dot{x}_1(t) = f_1\big(x_1(t), x_2(t), u(t)\big), \qquad \dot{x}_2(t) = f_2\big(x_1(t), x_2(t), u(t)\big).$$

A system **without input** ($\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x})$, as in Task 2a and 2b) is called **autonomous**.

7.2 From an n-th order ODE to state space

Many models arrive as one high-order differential equation, like the pendulum $ml^2\ddot{\theta} + b\dot{\theta} + mgl\sin\theta = 0$ or the I/O relation of Task 2(c). The recipe:

1. **Solve for the highest derivative.** For a second-order equation: $\ddot{y} = h(y, \dot{y}, u)$.
2. **Choose the states** as the output and its derivatives up to one order below the highest: $x_1 = y$, $x_2 = \dot{y}$, i.e. $\boldsymbol{x} = [x_1, x_2]^\top = [y, \dot{y}]^\top$ (for order n: $x_i = y^{(i-1)}$, $i = 1, \dots, n$).
3. **Write one first-order equation per state:**
   - $\dot{x}_1 = \dot{y} = x_2$ (this is just the definition of x_2),
   - $\dot{x}_2 = \ddot{y} = h(x_1, x_2, u)$ (the ODE, with $y \to x_1$, $\dot{y} \to x_2$).
4. **Output equation:** $y = x_1$, so $g(\boldsymbol{x}, u) = x_1$.

Pendulum example (slide 18). With $x_1 = \theta$ and $x_2 = \dot{\theta}$, the state-space equation is

$$\begin{bmatrix}\dot{x}_1 \\ \dot{x}_2\end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{b}{ml^2}x_2 - \frac{g}{l}\sin x_1\end{bmatrix}.$$

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8. Equilibrium points and how to find them

8.1 Definition 1.6 — equilibrium

For a nonlinear state-space model

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}),$$

an equilibrium point $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ satisfies

$$\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) = 0.$$

At an equilibrium point the state remains constant ($\dot{\boldsymbol{x}} = 0$): if the state starts at $\boldsymbol{x}_0$ and the input is held at $\boldsymbol{u}_0$, it stays at $\boldsymbol{x}_0$ for all future time. For an autonomous system the condition is $\boldsymbol{f}(\boldsymbol{x}_0) = 0$.

Finding equilibria means solving n (usually nonlinear) equations $f_1 = 0, \dots, f_n = 0$ for the n unknowns. There may be none, one, several, infinitely many isolated ones, or a whole continuum.

8.2 Strategy

1. **Look for an equation with only one unknown** (or that factorises) and solve it first; substitute into the others.
2. **Use trigonometric facts:**
   - $\sin z = 0 \iff z = k\pi$, $k \in \mathbb{Z}$.
   - $\cos(k\pi) = (-1)^k$ ($+1$ for even k, -1 for odd k); $\sin(k\pi) = 0$.
   - $|\sin z| \le 1$, $|\cos z| \le 1$; sin is odd, cos is even.
   - $1 - \cos z = 2\sin^2(z/2) \ge 0$, and $1 - \cos z \le z^2/2$.
3. **Always check the obvious candidate $\boldsymbol{x} = 0$.**
4. **Combine equations** (add, subtract) to cancel the hard terms.

5. **Bound the solutions** when the equations cannot be solved in closed form: e.g. if $a^2 + b^2 = -b \cos a$, then $a^2 + b^2 \leq |b|$, which confines (a, b) to a small disk. Inside a small region, signs of terms become known (e.g. $\cos a > 0$ for $|a| < \pi/2$), and this often rules out all but a few points.
6. **Verify** each candidate by substituting it back into **all** equations.
7. **For a system with input**, the equilibrium condition involves $\boldsymbol{u}_0$ too. Often one asks "for which constant $\boldsymbol{u}_0$ do equilibria exist?" (see Task 2c).

8.3 Examples from the lecture

- *Pendulum (slide 18):* the equilibrium points are given by $x_2 = 0$ and $\sin(x_1) = 0$. This leads to an infinite number of equilibrium points $(k\pi, 0)$, $k \in \mathbb{Z}$. Even k: hanging down; odd k: upright.
- *Cart-pendulum (slide 26):* $x_3 = x_4 = 0$ (velocities zero); with $u_0 = 0$ the third and fourth components of $\boldsymbol{f}$ need $\sin x_2 = 0$; the cart position x_1 is free. The lecture chooses $\boldsymbol{x}_0 = 0$, $u_0 = 0$.

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9. Linearization with Jacobians

9.1 Why do we linearize? (slide 15)

- Most real-world dynamical systems are nonlinear, and nonlinear systems are often difficult to analyse and control directly: no general closed-form solution, behaviour depending strongly on the initial condition, multiple equilibria, saturation.
- A linear model gives access to powerful tools: eigenvalues and modes, local stability analysis, controllability and observability, controller design (LQR, pole placement, observers).

A nonlinear system is not replaced by a linear system globally; it is approximated locally around an operating point.

9.2 One variable: the tangent line (slide 20)

A scalar function $f : \mathbb{R} \to \mathbb{R}$ is approximated around x_0 by the first two terms of the Taylor series:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

The neglected terms are of order $(x - x_0)^2$: the tangent line is a good approximation only for $x \approx x_0$, and the approximation error grows as we move away from the operating point. Example: $\sin x \approx x$ near 0 (since $\sin 0 = 0$, $\cos 0 = 1$) — the small-angle approximation of slide 23; $\sin x \approx -(x - \pi)$ near π (since $\cos \pi = -1$).

9.3 Several variables: partial derivatives and the Jacobian matrices (slide 19)

A **partial derivative** $\frac{\partial f_i}{\partial x_j}$ is the ordinary derivative of f_i with respect to x_j, treating all other variables as constants.

Rules you need (with x_2 treated as a constant when differentiating w.r.t. x_1, and vice versa):

Expression	$\frac{\partial}{\partial x_1}$	$\frac{\partial}{\partial x_2}$
$x_1^2 + x_2^2$	$2x_1$	$2x_2$
$x_2 \cos x_1$	$-x_2 \sin x_1$	$\cos x_1$

Expression	$\frac{\partial}{\partial x_1}$	$\frac{\partial}{\partial x_2}$
$x_1 \sin x_2$	$\sin x_2$	$x_1 \cos x_2$
$(1 + x_1)x_1 = x_1 + x_1^2$	$1 + 2x_1$	0
$(3 + x_2^2)x_2 = 3x_2 + x_2^3$	0	$3 + 3x_2^2$
$(1 + x_1 + x_1^2)\, u$	$(1 + 2x_1)\, u$	0

(Product rule: $(ab)' = a'b + ab'$; chain rule: $(\sin(h))' = \cos(h)\, h'$.) Tip: multiply out products before differentiating — fewer mistakes.

We can linearize $\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$ around the operating point $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ by expanding it into a Taylor series and cutting off all higher-order terms:

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \approx \boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} (\boldsymbol{x} - \boldsymbol{x}_0) + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} (\boldsymbol{u} - \boldsymbol{u}_0)$$

with the Jacobian matrices (row i belongs to f_i, column j to the variable)

$$\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \quad \text{and} \quad \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}} = \begin{bmatrix} \frac{\partial f_1}{\partial u_1} & \cdots & \frac{\partial f_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial u_1} & \cdots & \frac{\partial f_n}{\partial u_m} \end{bmatrix}.$$

9.4 The linearized state-space model (slides 21–22)

Consider the deviations from the equilibrium point $\Delta\boldsymbol{x} = \boldsymbol{x} - \boldsymbol{x}_0$ and $\Delta\boldsymbol{u} = \boldsymbol{u} - \boldsymbol{u}_0$. Since $\boldsymbol{x}_0$ is constant, $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_0 = 0$, and

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{x}) &= \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} - \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_0 \\ &\approx \left[\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} (\boldsymbol{x} - \boldsymbol{x}_0) + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} (\boldsymbol{u} - \boldsymbol{u}_0)\right] - \frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x}_0 \\ &= 0 + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} \Delta\boldsymbol{x} + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} \Delta\boldsymbol{u} - 0 \\ &= \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u}. \end{aligned}$$

The first 0 is $\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) = 0$ — this is why the equilibrium is the natural choice: **the resulting deviation model has no constant offset term** (slide 17). At a non-equilibrium operating point the constant $\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0)$ would remain.

If the system also has a time-invariant output equation $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u})$, then with $\Delta\boldsymbol{y} = \boldsymbol{y} - \boldsymbol{y}_0 = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u}) - \boldsymbol{g}(\boldsymbol{x}_0, \boldsymbol{u}_0)$:

$$\Delta\boldsymbol{y} \approx \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} \Delta\boldsymbol{x} + \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{u}}\right|_{(\boldsymbol{x}_0, \boldsymbol{u}_0)} \Delta\boldsymbol{u} = \boldsymbol{C}\Delta\boldsymbol{x} + \boldsymbol{D}\Delta\boldsymbol{u},$$

with the Jacobians $\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}} \in \mathbb{R}^{p \times n}$ and $\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{u}} \in \mathbb{R}^{p \times m}$ built like those of $\boldsymbol{f}$. Summary:

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}(\Delta \boldsymbol{x}) = \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u}, \quad \Delta\boldsymbol{y} = \boldsymbol{C}\Delta\boldsymbol{x} + \boldsymbol{D}\Delta\boldsymbol{u}, \quad \boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_0, \ \boldsymbol{B} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_0, \ \boldsymbol{C} = \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{x}}\right|_0, \ \boldsymbol{D} = \left.\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{u}}\right|_0}$$

9.5 Recipe

1. Write the model as $\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{x} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u})$, $\boldsymbol{y} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{u})$ (§7.2).
2. Find the equilibria $(\boldsymbol{x}_0, \boldsymbol{u}_0)$ from $\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) = 0$ (§8).
3. Compute the Jacobians **symbolically** (as functions of $\boldsymbol{x}$, $\boldsymbol{u}$).
4. **Evaluate** at each equilibrium ($\cdot|_0$) to get the numerical $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$. (Differentiate first, evaluate second — never the other way round.)
5. State the model in deviation variables, and say which equilibrium it belongs to.
6. Optionally interpret: eigenvalues of $\boldsymbol{A}$ (§10).

9.6 Lecture examples

Pendulum at $(k\pi, 0)$. $\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l}\cos x_1 & -\frac{b}{ml^2} \end{bmatrix}$, and $\cos(k\pi) = (-1)^k$:

$$\boldsymbol{A}_{\text{down}} = \begin{bmatrix} 0 & 1 \\ -\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \ (k \text{ even}), \qquad \boldsymbol{A}_{\text{up}} = \begin{bmatrix} 0 & 1 \\ +\frac{g}{l} & -\frac{b}{ml^2} \end{bmatrix} \ (k \text{ odd}).$$

For the undamped pendulum of slide 23 ($b = 0$, written with θ and $\omega = \dot\theta$: $\frac{\mathrm{d}}{\mathrm{d}t}\theta = \omega$, $\frac{\mathrm{d}}{\mathrm{d}t}\omega = -\frac{g}{l}\sin\theta$), the small-angle approximation $\sin\theta \approx \theta$ gives exactly $\boldsymbol{A}_{\text{down}}$:

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix}\Delta\theta \\ \Delta\omega\end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -g/l & 0 \end{bmatrix}\begin{bmatrix}\Delta\theta \\ \Delta\omega\end{bmatrix}.$$

Cart-pendulum (slides 24–26). $\boldsymbol{x}(t)^\top = [\zeta(t), \phi(t), \dot\zeta(t), \dot\phi(t)]$, $u(t) = F(t)$, $y(t) = x_2(t)$. At $x_2 = 0$: $\sin x_2 = 0$, $\cos x_2 = 1$, and every term with x_4^2 has zero derivative at $x_4 = 0$ (derivative $2x_4 = 0$). With $\boldsymbol{x}_0 = 0$, $u_0 = 0$:

$$\boldsymbol{A} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & \frac{mg}{M} & 0 & 0 \\ 0 & -\frac{(M+m)g}{lM} & 0 & 0 \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} 0 \\ 0 \\ \frac{1}{M} \\ -\frac{1}{lM} \end{bmatrix}, \quad \boldsymbol{C} = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{D} = 0.$$

9.7 Validity

The linear approximation is only valid **locally** around the operating point and works well when $\Delta\boldsymbol{x}$ and $\Delta\boldsymbol{u}$ are sufficiently small (slide 21). Each equilibrium has its *own* linear model, and they can behave completely differently (pendulum down: oscillation; up: falls over). The figures in `python/figures/` show nonlinear and linearized trajectories lying on top of each other near the equilibrium and separating further away.

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10. What the linear model says about stability (2×2 toolkit)

Task 2 asks only for the linearization, but interpreting it is what linearization is for, and you need eigenvalues for Task 3 anyway.

10.1 Eigenvalues of a 2×2 matrix

The eigenvalues λ of $\boldsymbol{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ solve the **characteristic equation**

$$\det(\lambda \boldsymbol{I} - \boldsymbol{A}) = (\lambda - a_{11})(\lambda - a_{22}) - a_{12}a_{21} = \lambda^2 - \operatorname{tr}(\boldsymbol{A})\,\lambda + \det(\boldsymbol{A}) = 0,$$

with $\operatorname{tr} \boldsymbol{A} = a_{11} + a_{22}$ and $\det \boldsymbol{A} = a_{11}a_{22} - a_{12}a_{21}$. So

$$\lambda_{1,2} = \frac{\operatorname{tr} \boldsymbol{A} \pm \sqrt{(\operatorname{tr} \boldsymbol{A})^2 - 4 \det \boldsymbol{A}}}{2}, \qquad \lambda_1 + \lambda_2 = \operatorname{tr} \boldsymbol{A}, \quad \lambda_1 \lambda_2 = \det \boldsymbol{A}.$$

For a **diagonal or triangular** matrix, the eigenvalues are simply the diagonal entries.

10.2 Continuous time: sign of the real parts

For $\frac{\mathrm{d}}{\mathrm{d}t}(\Delta \boldsymbol{x}) = \boldsymbol{A}\Delta \boldsymbol{x}$ the solution is built from terms $e^{\lambda t}$ (week 3). Hence:

Eigenvalues of $\boldsymbol{A}$	Behaviour of $\Delta \boldsymbol{x}$	Name (2D)
all $\operatorname{Re} \lambda < 0$	decays to 0	stable node / focus
some $\operatorname{Re} \lambda > 0$	grows	unstable (saddle if the other is < 0)
some $\operatorname{Re} \lambda = 0$, rest < 0	neither decays nor necessarily grows	marginal case

Quick 2×2 test: **both eigenvalues have negative real part ⇔ $\operatorname{tr} \boldsymbol{A} < 0$ and $\det \boldsymbol{A} > 0$.** $\det \boldsymbol{A} < 0$ ⇔ real eigenvalues of opposite sign ⇔ saddle.

10.3 Linearization and local stability of the nonlinear system (preview of week 9)

Lyapunov's indirect method: if all eigenvalues of $\boldsymbol{A} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\big|_0$ have $\operatorname{Re} \lambda < 0$, the equilibrium of the *nonlinear* system is locally asymptotically stable; if at least one has $\operatorname{Re} \lambda > 0$, it is unstable. If some eigenvalue has $\operatorname{Re} \lambda = 0$ (and none positive), the linearization is **inconclusive** — the neglected higher-order terms decide. Task 2(c) is such a case.

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11. Discrete-time LTI systems: solution of the state equation

11.1 The model

$$\boldsymbol{x}[k+1] = \boldsymbol{A}\boldsymbol{x}[k] + \boldsymbol{B}u[k], \qquad \boldsymbol{y}[k] = \boldsymbol{C}\boldsymbol{x}[k] + \boldsymbol{D}u[k], \qquad k \in \mathbb{Z}.$$

Unlike a differential equation, a difference equation is a **recipe you can run**: given $\boldsymbol{x}[0]$ and the inputs, compute $\boldsymbol{x}[1]$, then $\boldsymbol{x}[2]$, and so on.

Reminder of matrix–vector multiplication: $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} p \\ q \end{bmatrix} = \begin{bmatrix} a_{11}p + a_{12}q \\ a_{21}p + a_{22}q \end{bmatrix}$ (row times column).

11.2 Free response ($u[k] = 0$ for all k)

$\boldsymbol{x}[1] = \boldsymbol{A}\boldsymbol{x}[0]$, $\boldsymbol{x}[2] = \boldsymbol{A}\boldsymbol{x}[1] = \boldsymbol{A}^2\boldsymbol{x}[0]$, ..., so

$$\boldsymbol{x}[k] = \boldsymbol{A}^k\boldsymbol{x}[0], \qquad \boldsymbol{A}^0 = \boldsymbol{I}.$$

11.3 Full response — derivation

Iterate with the input:

$$\begin{aligned}
\boldsymbol{x}[1] &= \boldsymbol{A}\boldsymbol{x}[0] + \boldsymbol{B}u[0] \\
\boldsymbol{x}[2] &= \boldsymbol{A}\boldsymbol{x}[1] + \boldsymbol{B}u[1] = \boldsymbol{A}^2\boldsymbol{x}[0] + \boldsymbol{A}\boldsymbol{B}u[0] + \boldsymbol{B}u[1] \\
\boldsymbol{x}[3] &= \boldsymbol{A}\boldsymbol{x}[2] + \boldsymbol{B}u[2] = \boldsymbol{A}^3\boldsymbol{x}[0] + \boldsymbol{A}^2\boldsymbol{B}u[0] + \boldsymbol{A}\boldsymbol{B}u[1] + \boldsymbol{B}u[2].
\end{aligned}$$

The pattern: the input applied at step i has been "processed" by $\boldsymbol{A}$ exactly $k-1-i$ times by step k. Hence

$$\boxed{\boldsymbol{x}[k] = \underbrace{\boldsymbol{A}^k\boldsymbol{x}[0]}_{\text{free (zero-input) response}} + \underbrace{\sum_{i=0}^{k-1} \boldsymbol{A}^{k-1-i}\boldsymbol{B}\,u[i]}_{\text{forced (zero-state) response}}}, \qquad k \geq 1.$$

Proof by induction. For $k = 1$ it reads $\boldsymbol{x}[1] = \boldsymbol{A}\boldsymbol{x}[0] + \boldsymbol{B}u[0]$ ✓. If it holds for k, then $\boldsymbol{x}[k+1] = \boldsymbol{A}\boldsymbol{x}[k] + \boldsymbol{B}u[k] = \boldsymbol{A}^{k+1}\boldsymbol{x}[0] + \sum_{i=0}^{k-1} \boldsymbol{A}^{k-i}\boldsymbol{B}u[i] + \boldsymbol{A}^0\boldsymbol{B}u[k] = \boldsymbol{A}^{k+1}\boldsymbol{x}[0] + \sum_{i=0}^{k} \boldsymbol{A}^{k-i}\boldsymbol{B}u[i]$ ✓.

This is exactly the split $\boldsymbol{y}(\boldsymbol{x}_0, \boldsymbol{u}, t) = \boldsymbol{y}(0, \boldsymbol{u}, t) + \boldsymbol{y}(\boldsymbol{x}_0, 0, t)$ of Definition 1.1 (§2.1). The sum is a **discrete convolution** of the input with $\boldsymbol{A}^j\boldsymbol{B}$ (week 3).

11.4 Impulse response

With $\boldsymbol{x}[0] = 0$ and the unit impulse $u[k] = \delta[k]$ (1 for $k = 0$, 0 for $k \neq 0$), only the $i = 0$ term survives:

$$\boldsymbol{x}[k] = \boldsymbol{A}^{k-1}\boldsymbol{B}\ (k \geq 1), \quad \boldsymbol{x}[0] = 0, \qquad \boldsymbol{y}[k] = \boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B}\ (k \geq 1), \quad \boldsymbol{y}[0] = \boldsymbol{D}.$$

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12. Computing A^k with eigenvalues and eigenvectors

12.1 Eigenvalues and eigenvectors

$\boldsymbol{v} \neq 0$ is an **eigenvector** of $\boldsymbol{A}$ with **eigenvalue** λ if

$$\boldsymbol{A}\boldsymbol{v} = \lambda\boldsymbol{v}.$$

$\boldsymbol{A}$ acts on $\boldsymbol{v}$ just by scaling it. Applying $\boldsymbol{A}$ k times: $\boldsymbol{A}^k\boldsymbol{v} = \lambda^k\boldsymbol{v}$. **This is the whole trick.**

How to compute them:

1. Eigenvalues: solve $\det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0$ (§10.1).

2. For each λ: solve $(\lambda \boldsymbol{I} - \boldsymbol{A})\boldsymbol{v} = 0$ (the same matrix as in step 1; $(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = 0$ is equivalent — every sign flips, the solutions don't change). The matrix $\lambda \boldsymbol{I} - \boldsymbol{A}$ is singular, so its rows are multiples of each other; one row $\begin{bmatrix} p & q \end{bmatrix}$ gives $pv_1 + qv_2 = 0$, and $\boldsymbol{v} = \begin{bmatrix} q \\ -p \end{bmatrix}$ (or any multiple) works.
3. Check: $\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}$.

12.2 Method 1 — decompose the initial state

If $\boldsymbol{A} \in \mathbb{R}^{2\times 2}$ has two distinct eigenvalues $\lambda_1 \neq \lambda_2$ with eigenvectors $\boldsymbol{v}_1, \boldsymbol{v}_2$, these form a basis: every $\boldsymbol{x}[0]$ can be written as $\boldsymbol{x}[0] = c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2$ (solve a 2×2 linear system for c_1, c_2). Then

$$\boldsymbol{x}[k] = \boldsymbol{A}^k \boldsymbol{x}[0] = c_1 \lambda_1^k \, \boldsymbol{v}_1 + c_2 \lambda_2^k \, \boldsymbol{v}_2.$$

12.3 Method 2 — diagonalization

Put the eigenvectors as columns in $\boldsymbol{V} = [\boldsymbol{v}_1 \ \boldsymbol{v}_2]$ and $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \lambda_2)$. Then $\boldsymbol{A}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda}$, so $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}$ and

$$\boldsymbol{A}^k = (\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1})(\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1})\cdots(\boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1}) = \boldsymbol{V}\boldsymbol{\Lambda}^k\boldsymbol{V}^{-1}, \qquad \boldsymbol{\Lambda}^k = \begin{bmatrix} \lambda_1^k & 0 \\ 0 & \lambda_2^k \end{bmatrix}.$$

(All inner $\boldsymbol{V}^{-1}\boldsymbol{V} = \boldsymbol{I}$ cancel.) **2×2 inverse:** $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}^{-1} = \dfrac{1}{a_{11}a_{22} - a_{12}a_{21}} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix}$.

Equivalently, with $\boldsymbol{w}_i^\top$ the rows of $\boldsymbol{V}^{-1}$: $\boldsymbol{A}^k = \lambda_1^k \, \boldsymbol{v}_1\boldsymbol{w}_1^\top + \lambda_2^k \, \boldsymbol{v}_2\boldsymbol{w}_2^\top$ (sum of rank-one "mode" matrices).

Always **check** a closed form at $k = 0$ (must give $\boldsymbol{I}$ or $\boldsymbol{x}[0]$) and $k = 1$ (must give $\boldsymbol{A}$ or $\boldsymbol{A}\boldsymbol{x}[0]$).

12.4 Powers of negative numbers

$(-1)^k$ alternates $1, -1, 1, -1, \ldots$ (bounded, never converges). $(-2)^k$ alternates sign and doubles in size: $1, -2, 4, -8, 16, \ldots$ In general $|\lambda^k| = |\lambda|^k$.

12.5 A useful shortcut

If $\boldsymbol{B}$ (or $\boldsymbol{x}[0]$) happens to be an eigenvector, i.e. $\boldsymbol{A}\boldsymbol{B} = \lambda\boldsymbol{B}$, then $\boldsymbol{A}^j\boldsymbol{B} = \lambda^j\boldsymbol{B}$ for every j — no matrix powers needed. Look for this before computing (Task 3 c, d).

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13. Modes, internal stability and BIBO stability

13.1 Modes in discrete time

From §12.2, the free response is a sum of **modes** $c_i\lambda_i^k\boldsymbol{v}_i$. Each mode behaves according to its eigenvalue:

λ (real)	λ^k	mode
$0 < \lambda < 1$	$\to 0$ monotonically	decays
$-1 < \lambda < 0$	$\to 0$ with alternating sign	decays, oscillating

λ (real)	λ^k	mode
$\lambda = 1$	1	constant
$\lambda = -1$	± 1 alternating	bounded, does **not** converge
$\lambda > 1$	$\to \infty$	diverges
$\lambda < -1$	$\pm\infty$ alternating	diverges, oscillating

(For complex λ, $|\lambda|$ decides the growth and the angle gives a rotation.) **The boundary in discrete time is the unit circle** $|\lambda| = 1$, whereas in continuous time it is the imaginary axis $\operatorname{Re}\lambda = 0$.

13.2 Internal stability

The exercise sheet defines it as the behaviour of the state as $k \to \infty$ in the absence of an external input, i.e. of $\boldsymbol{x}[k] = \boldsymbol{A}^k \boldsymbol{x}[0]$:

- **Asymptotically stable:** $\boldsymbol{x}[k] \to 0$ for every $\boldsymbol{x}[0]$ ⇔ all $|\lambda_i| < 1$.
- **Marginally stable:** $\boldsymbol{x}[k]$ bounded for every $\boldsymbol{x}[0]$ — requires all $|\lambda_i| \leq 1$ (and eigenvalues on the circle not "repeated with a Jordan block" — week 9).
- **Unstable:** some $\boldsymbol{x}[0]$ makes $\boldsymbol{x}[k]$ unbounded ⇔ some $|\lambda_i| > 1$ (or a defective eigenvalue on the circle).

For a **specific** $\boldsymbol{x}[0]$, convergence depends on which modes it excites: if $c_i = 0$ for every bad eigenvalue, those modes never appear. The answer to "does it converge for $\boldsymbol{x}[0] = \ldots$?" therefore needs the decomposition $\boldsymbol{x}[0] = \sum c_i \boldsymbol{v}_i$.

13.3 BIBO stability

BIBO stability is about the input–output behaviour from $\boldsymbol{x}[0] = 0$: every bounded input must give a bounded output. For an LTI system it holds if and only if the impulse response is absolutely summable, $\sum_k \|\boldsymbol{C}\boldsymbol{A}^{k-1}\boldsymbol{B}\| < \infty$ (week 10). In particular: **if the response to the bounded input $\delta[k]$ is unbounded, the system is not BIBO stable.**

Internal vs BIBO. Internal (asymptotic) stability implies BIBO stability, but not conversely: an unstable mode that the input cannot excite (or the output cannot see) leaves the I/O behaviour bounded. In Task 3 both fail, because $\boldsymbol{B}$ lies exactly on the unstable eigenvector.

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14. Cheat sheet

System properties (I/O relationship $y(t) = \ldots$)

Property	Test (lecture definition)	Quick signs of failure
Linear (Def. 1.1)	$u_1 \to y_1,\ u_2 \to y_2 \Rightarrow \alpha_1 u_1 + \alpha_2 u_2 \to \alpha_1 y_1 + \alpha_2 y_2$; first check $u = 0 \Rightarrow y = 0$	u^2, $x(t)x(t-1)$, $\cos(u)$, e^u, constant offsets
Time-invariant (Def. 1.2)	$y(u(t), t{-}T) = y(u(t{-}T), t)$; on the left replace **every** t by $t - T$	explicit t outside u (e.g. $\cos 2\pi f_0 t$), time scaling $u(2t)$, reversal $x[4-n]$
Causal (Def. 1.3)	$u_1 = u_2\ \forall t \leq T \Rightarrow y_1 = y_2\ \forall t \leq T$; all read times $\leq t$	$u(t+1)$, $u(2t)$, $x[4-n]$
BIBO stable	$\lvert u\rvert \leq M \Rightarrow \lvert y\rvert \leq M'$	accumulation, e^t, $t\,u(t)$

Memoryless ⇒ causal. Time-only factors ($c(t)u(t)$) keep linearity. Changing only time arguments keeps linearity.

Linearization

$$\boldsymbol{f}(\boldsymbol{x}_0, \boldsymbol{u}_0) = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\Delta\boldsymbol{x}) = \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u}, \quad \Delta\boldsymbol{y} = \boldsymbol{C}\Delta\boldsymbol{x} + \boldsymbol{D}\Delta\boldsymbol{u}, \qquad \boldsymbol{A} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}}\right|_0, \boldsymbol{B} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}}\right|_0, \boldsymbol{C} = \frac{\partial}{\partial}$$

Differentiate first, evaluate second. $\sin(k\pi) = 0$, $\cos(k\pi) = (-1)^k$. 2×2: $\lambda^2 - \operatorname{tr}\boldsymbol{A}\,\lambda + \det\boldsymbol{A} = 0$; stable ⇔ $\operatorname{tr}\boldsymbol{A} < 0$, $\det\boldsymbol{A} > 0$; saddle ⇔ $\det\boldsymbol{A} < 0$.

Discrete time

$$\boldsymbol{x}[k] = \boldsymbol{A}^k\boldsymbol{x}[0] + \sum_{i=0}^{k-1}\boldsymbol{A}^{k-1-i}\boldsymbol{B}u[i], \qquad \boldsymbol{A}^k = \boldsymbol{V}\boldsymbol{\Lambda}^k\boldsymbol{V}^{-1}, \qquad \boldsymbol{A}^k\boldsymbol{v} = \lambda^k\boldsymbol{v}.$$

Asymptotically stable ⇔ all $|\lambda| < 1$. Impulse response $\boldsymbol{x}[k] = \boldsymbol{A}^{k-1}\boldsymbol{B}$; unbounded ⇒ not BIBO stable.